

Pharmaceutical Sales Analytics Dashboard Project Description

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Track:

Data Analysis using Power BI

Project Overview

The Pharmaceutical Sales Analytics Dashboard is a Business Intelligence solution developed using Microsoft Power BI to transform raw pharmaceutical sales data into interactive dashboards that support strategic decision-making.

The project provides decision-makers with a comprehensive view of sales performance, profitability, pharmacy performance, product analysis, and business opportunities through dynamic reports, KPIs, and interactive visualizations.

By integrating data modeling, DAX calculations, and business intelligence techniques, the dashboard enables organizations to monitor performance efficiently and identify opportunities for growth.

Business Problem

Pharmaceutical companies generate large volumes of sales transactions every day across different pharmacies, products, and regions. Although this data contains valuable business insights, it is often stored in spreadsheets that are difficult to analyze manually.

As a result, managers face several challenges:

- Difficulty monitoring overall sales performance.

- Limited visibility into product profitability.

- Inability to compare pharmacy performance.

- Time-consuming manual reporting.

- Difficulty identifying low-performing products and business opportunities.

Without an interactive reporting solution, decision-making becomes slower and less effective.

Project Objectives

The main objectives of this project are to:

- Develop an interactive Power BI dashboard for pharmaceutical sales analysis.
- Monitor key business performance indicators (KPIs).
- Analyze revenue, costs, and profit margins.
- Compare pharmacy performance across different regions.
- Evaluate product profitability.
- Identify low-performing products.
- Support management with data-driven insights.
- Improve business reporting efficiency.

Dataset Description

The project is based on a pharmaceutical sales dataset containing transactional sales information collected from multiple pharmacies.

The dataset includes:

Data Structure Data Model

The project follows a Star Schema data model.

Fact Table

T_FactSales

Contains:

Revenue
Cost
Margin
Units Sold
Promotion Flag

Dimension Tables

T_DimProduct

Product Name
Brand
Category

T_DimPharmacy

Pharmacy
Region

T_DimDate

Date
Month
Quarter
Year

Medicine_Details

Medicine Description
Manufacturer
Medical Information

The data model follows the Star Schema approach to improve query performance and simplify relationships between tables.

Technologies Used

The project was developed using the following technologies:

- Microsoft Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Microsoft Excel
- Data Modeling (Star Schema)
- Business Intelligence Techniques

Dashboard Features

The dashboard consists of four interactive pages:

Executive Dashboard

Provides an overview of:

Total Revenue

Total Margin

Total Cost

Profit Margin %

Revenue Trend

Sales Performance

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Revenue Trend

Sales Performance

Medication Details

Displays detailed information about pharmaceutical products and medicine descriptions.

Product & Pharmacy Analytics

Provides insights into:

- Product Performance
- Pharmacy Performance
- Regional Analysis
- Best and Worst Performing Products

Business Problems & Opportunities

Highlights:

- Low Profit Products
- Promotion Effectiveness
- Business Risks
- Revenue Opportunities
- Business Recommendations

Key Performance Indicators (KPIs)

The dashboard includes several business KPIs, including:

- Total Revenue
- Total Cost
- Total Margin
- Margin Percentage
- Revenue per Pharmacy
- Product Performance
- Promotion Analysis
- Sales Trend
- Business Opportunity Indicators

Business Insights

The dashboard enables decision-makers to:

- Monitor business performance in real time.
- Identify high-performing products.
- Detect low-profit products.
- Compare pharmacy performance.
- Evaluate promotional campaigns.
- Discover new business opportunities.
- Improve strategic decision-making.

Future Enhancements

Future improvements may include:

- AI-powered forecasting.
- Sales prediction models.
- Automatic data refresh.
- Power BI Service deployment.
- Mobile dashboard optimization.
- Integration with SQL Server.
- Advanced machine learning analytics.

Project Deliverables

The final project includes:

- Power BI Dashboard (.pbix)
- Interactive Reports
- Project Documentation
- PowerPoint Presentation
- Dataset
- DAX Measures
- Source Code
- Business Analysis Report

Conclusion

The Pharmaceutical Sales Analytics Dashboard successfully demonstrates how Business Intelligence can transform raw pharmaceutical sales data into meaningful insights. Through interactive visualizations, KPIs, and data-driven analysis, the project helps decision-makers monitor performance, improve profitability, and identify strategic business opportunities.

This project also strengthened practical skills in Power BI, Power Query, DAX, Data Modeling, and Business Intelligence, while providing a scalable foundation for future analytical and predictive solutions.